Fine-Tuning Medical Vision Language Models for Pleural Effusion Classification in Female Subpopulations

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Abstract

Underrepresented demographic subgroups and label scarcity are two major issues that deep learning applications face in clinical settings (Seyyed-Kalantari et al., 2021). Artificial Intelligence systems have proven to underperform when tasked with predicting diagnoses for subpopulations defined by age, gender, race, and socioeconomic status (Stogiannos et al., 2025). To address this challenge, this paper fine-tunes BioViL, a medical vision-language model pretrained on chest X-rays and radiology reports from MIMIC-CXR, on the CheXpert Plus dataset (Chambon et al., 2024). We constrain the dataset to female patients, targeting pleural effusion classification as an underexplored subgroup-specific task. Low-Rank Adaptation (LoRA) is applied separately to the text and image encoder, and to both simultaneously, to yield three distinct modality conditions for an ablation study. Label budgets are varied across N = 25, 50, 100, 200, 500, and full. Our central hypothesis is that multimodal fusion provides disproportionately large gains at low label budgets, where radiology text would compensate for insufficient labelled image signal. We find no evidence to support this hypothesis: across all three modalities and all low-budget conditions (N = 25 to 500), test AUROC remains near random chance ($\sim$0.50), and multimodal fusion does not outperform image-only or text-only baselines at any low budget. At full budget, performance improves only marginally (multimodal AUROC 0.523, text-only LoRA 0.522), with the two modalities tracking each other closely. This indicates that the combined effects of demographic restriction, label noise from NLP-derived ground truth, and the masking of diagnostic terms create a learning regime in which neither modality choice nor budget alone recovers strong classification performance. We argue that richer textual inputs which retain diagnostic context while preventing label leakage are needed to test the multimodal advantage hypothesis fairly.

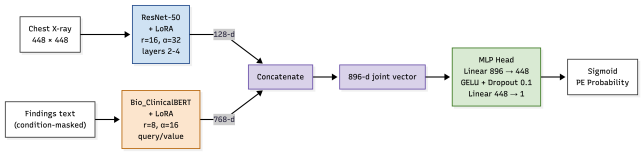


Figure 1: Multimodal BioViL architecture with LoRA adapters on both encoders. The image and text vectors are concatenated and passed through a two-layer MLP head to produce the pleural effusion probability.

1. Introduction

Current research consensus in the medical community suggests that optimising for aggregate performance metrics systematically disadvantages specific patient subgroups, as high overall accuracy can mask substantial performance gaps for underrepresented populations. Seyyed-Kalantari et al. (2021) demonstrated this directly, showing that advanced chest X-ray classifiers underdiagnose female patients, Black patients, and socioeconomically marginalised individuals. The underlying nature of the models is to maximise average performance, hence, it typically acquires biases that may not be clearly observed in predictive outputs. This stems from skewed training distributions which produce systematic false negatives in diagnosing specific patient subgroups. A strategy that can be employed to potentially resolve this underlying issue is subgroup specific fine-tuning. By limiting the labelled fine-tuning set to the target subpopulation, the model can precisely be refined for that underlying distribution. Restricting the number of labelled examples seen by the model during training is crucial to the fine-tuning process. That being said, there is an inherent issue with this approach that may lead to overfitting the model that must be considered. A demographically constrained dataset is smaller, hence, fine-tuning a large pretrained model on these limited labels may cause the model to generalize quite poorly on test data that is previously unseen. That is why LoRA is a clear solution to the issue outlined above because it involves parameter-efficient fine-tuning, by sharply decreasing the amount of trainable parameters (Hu et al., 2021).

Modality poses an additional concern for answering this research question. Therefore, understanding why BioViL is the right model for this experiment is crucial. Its main strength is that it is a medical vision-language model pretrained on paired image-report data. Additionally, it upholds visual and textual representations of clinical knowledge (Bannur et al., 2023). We use CheXpert Plus, a large-scale chest X-ray dataset released by Stanford University's AI Medicine and Imaging Institute that pairs radiograph images with free-text radiology reports (Chambon et al., 2024). A critical part of this study was to rigorously test whether the text modality contributes meaningful signal beyond image-only fine-tuning while simultaneously preventing label leakage from impression text into training inputs. By artificially constraining the number of labels in training, as described above, the radiology text provides a robust signal that counterbalances the limited visual evidence available to the classifier. Looking

at varying levels of label budgets allows the experiment to capture the benefit of multimodal fusion at differing scales. It also serves the purpose of identifying whether a crossover persists at which image-only performance could match multimodal performance, which is an empirically open-ended question that this report tries to tackle. To ensure that the model learns from descriptive visual language rather than memorising the conclusion given by the doctor in diagnosis, the radiology findings text was used with explicit masking of pleural effusion terms.

An inherent gap in previous work is that it has not been shown whether training regime choices can possibly mitigate model biases. Seyyed-Kalantari et al. (2021) did not provide a solution to the systematic underdiagnosing of female patients (among other subgroups) on CheXpert-scale data. Neither did the researchers propose an experimental design with varying modality choices like the experimental design of this paper. Therefore, examining the interaction between multimodal fine-tuning and label budget scarcity in a demographically constrained subpopulation addresses a direct gap in the research. This setup characterises whether descriptive text provides a useful supervisory signal at low N and whether the benefit diminishes as visual evidence grows sufficient on its own. This paper makes three contributions. First, it characterises modality choice and label budget interactions in the low-data regime for pleural effusion classification on CheXpert Plus. Second, it proposes a demographically constrained fine-tuning setting driven by recorded underdiagnosis bias against female patients, examining whether subgroup-specific fine-tuning produces quantifiable improvements over a general baseline evaluated on the same female test set. Third, by contrasting the image encoder, text encoder, and multimodal encoder with LoRA employed on each architecture, it provides an encoder-level ablation study that selectively adapts a pretrained medical vision-language model via low-rank adaptation.

Across all three modality conditions and six label budgets, fine-tuned models perform near random chance (AUROC $\sim$0.50). Marginal improvement is observed only at the full budget condition, where the multimodal encoder achieves AUROC 0.523 and the text-only LoRA encoder achieves 0.522, essentially identical performance. The hypothesised disproportionate gains from multimodal fusion at low label budgets did not materialise: at no tested low budget (N=25 to N=500) does the multimodal encoder outperform the text-only or image-only baselines. Combining demographic restriction, label noise derived from NLP-extracted ground truth, and masked text input generates a learning regime where neither modality choice nor label budget alone recovers strong pleural effusion classification performance. The central hypothesis therefore needs to be revisited under a different methodological framework, possibly with cleaner label sources and richer text inputs that retain diagnostic context while still preventing label leakage. A practical limitation in this field is the difficulty of obtaining such richer text annotations, which would require systematic engagement of medical professionals during data preparation. These findings extend Seyyed-Kalantari et al. (2021)'s inference-time bias results into the fine-tuning regime, suggesting that subgroup-specific fine-tuning alone is not sufficient to close the demographic performance gap on this classification task.

The remainder of the paper is structured as follows: Section 2 reviews related work; Section 3 describes the architecture; Section 4 covers training; Section 5 presents results; and Section 6 concludes.

2. Related Work

2.1 Algorithmic Bias in Medical Imaging

The foundational work of Seyyed-Kalantari et al. (2021) demonstrated that state-of-the-art chest X-ray classifiers trained on large public datasets, including CheXpert and MIMIC-CXR, exhibit systematic underdiagnosis of subgroups. A subsequent analysis by Seyyed-Kalantari et al. (2023) directly quantified these disparities in a chest radiography foundation model, finding that performance decreased by 6.8–7.8% for female patients on the "no finding" label. Notably, pleural effusion exhibited among the largest demographic performance gaps of any pathology examined, providing direct empirical motivation for targeting this condition in a female-constrained fine-tuning study.

Obermeyer and Topol (2024) cautioned that algorithmic corrections to demographic shortcuts can produce locally optimal models that fail to generalise to new populations, suggesting that subgroup-specific fine-tuning must be accompanied by rigorous cross-cohort evaluation. This motivates our inclusion of a mixed-sex general baseline trained on the full cohort and evaluated on the female subgroup as a comparison condition, allowing us to assess whether demographic-specific fine-tuning produces quantifiable gains over population-level training.

2.2 Medical Vision-Language Models

BioViL (Bannur et al., 2023), the backbone model for this study, is a dual-encoder vision-language model pretrained on paired chest X-rays and radiology reports from MIMIC-CXR using contrastive learning objectives. Its domain-adapted CXR-BERT text encoder is specifically equipped to handle the specialised language of clinical reports, including negations and uncertainty markers that general-purpose language models struggle with.

Shurrab et al. (2025) found that multimodal models outperform unimodal baselines in zero-shot settings, but that this advantage narrows substantially after fine-tuning. This raises the empirical question of whether multimodal fusion provides any consistent benefit under the label-scarce, demographically

constrained conditions studied here.

Zhang et al. (2024) evaluated five vision-language foundation models across five chest X-ray datasets and found that these models consistently underdiagnose marginalised groups compared to board-certified radiologists, with the largest disparities seen in intersectional subgroups such as Black female patients. However, their study did not investigate whether targeted fine-tuning strategies could reduce these disparities, a gap the present study aims to address.

2.3 Parameter-Efficient Fine-Tuning

Fine-tuning large pretrained models in clinical settings is challenging due to computational cost, risk of catastrophic forgetting, and overfitting under label scarcity. Parameter-efficient fine-tuning (PEFT) methods address these issues by updating only a small fraction of model parameters while keeping pretrained weights frozen.

LoRA (Hu et al., 2021) is the primary adaptation technique in this study. Rather than updating full weight matrices, LoRA introduces low-rank decompositions that drastically reduce the number of trainable parameters, typically by a factor of 10,000 for large transformer layers. Zhu et al. (2023) confirmed that LoRA outperforms full fine-tuning for medical image diagnosis tasks, particularly in preserving pretrained representations, which is critical when fine-tuning on a demographically constrained dataset where the risk of over-specialisation is high. Prior work has also suggested that optimal LoRA rank should scale with the amount of available labelled data, with small ranks preferable at very low N to reduce overfitting risk.

Prompt tuning (Lester et al., 2021) offers an alternative approach, prepending learnable token embeddings to a frozen model's input to steer its representations without modifying any weights. Shakeri et al. (2024) found that prompt-based and adapter-based methods perform comparably at low shot counts for chest X-ray classification, while adapter-based methods gain an edge as more labelled data becomes available. Rahman et al. (2026) proposed that combining LoRA with prompt tuning yields improvements beyond either method in isolation, motivating the investigation of whether this holds in the demographically constrained setting studied here.

3. Architecture

3.1 PyTorch and BioViL

The three models were implemented in PyTorch (Paszke et al., 2019), which offers native integration with HuggingFace Transformers and PEFT without the need for third-party wrappers. PyTorch's dynamic computation graphs simplify the implementation of modality-conditional forward passes, where the active encoder pathway changes between image-only, text-only, and multimodal conditions. The framework also supports specialised medical imaging libraries, including the official BioViL implementation in `health_multimodal` (Microsoft Research, 2023).

BioViL (Bannur et al., 2023) was used as the base model in the study. Given its dual-encoder architecture, it has two separate neural networks that run simultaneously, one for image and one for text. The image encoder is called an image tower and the text encoder is called a text tower. In this specific setting, the image tower employed is a ResNet-50 pretrained on chest X-rays, which runs the input through convolutional layers and outputs 128-dimensional vector representations. These global embeddings portray the visual content of the image. The text tower is Bio_ClinicalBERT (Alsentzer et al., 2019), a BERT language model pretrained on clinical notes that generates a 768-dimensional representation. In the multimodal setting, both representations are concatenated to construct an 896-dimensional merged vector prior to the classification head.

An additional neural network is added on top of the encoders for the final prediction. In the multimodal condition it sits on top of the 896-dimensional concatenated vector, and similarly the text-only and image-only conditions feed their respective vectors into the same head architecture. The head is a two-layer multilayer perceptron that projects a single unnormalised score (a logit). Training uses binary cross-entropy loss, with a sigmoid activation applied at inference time so probabilities are interpretable between 0 and 1. Figure 1 displays the full architecture after the abstract.

3.2 LoRA Adapters

Low-Rank Adaptation (LoRA) is implemented with modality-appropriate hyperparameters. The image encoder uses rank $r = 16$, scaling factor $\alpha = 32$, and dropout 0.1. The text encoder uses rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.1. In the multimodal configuration, both LoRA adapters operate jointly: the image branch retains its rank $r = 16$, $\alpha = 32$, and dropout 0.1 settings, while the text branch uses rank $r = 8$, $\alpha = 16$, and dropout 0.05. The lower text dropout in the multimodal configuration reduces overfitting risk under joint optimisation, where the model can otherwise rely too heavily on the text branch given the image branch's larger LoRA capacity. Higher rank in the image encoder reflects the greater representational adaptation required for visual features in a demographically constrained subgroup, while a lower rank in the text encoder reduces overfitting risk at low label budgets (Hu et al., 2021). Dropout randomly zeroes out a fraction of LoRA activations during training to mitigate overfitting.

The image encoder's adapters target the 3×3 convolutional projections within layers 2, 3, and 4 of ResNet-50, rather than the earliest layers. This targets the intermediate and high-level stages where features become more relevant to pathology recognition, while leaving low-level edge and texture extraction unchanged. For each targeted layer, a low-rank

bypass is added: a convolution reduces the channel dimension to rank $r = 16$, followed by a 1×1 convolution projecting back to the original output channels. The bypass matrix is initialised to zero so training begins from the pretrained representation, and batch normalisation layers are kept in evaluation mode throughout to preserve pretrained running statistics under small training sets. Within the text encoder, LoRA adapters are injected into the query and value projections of every self-attention layer of Bio_ClinicalBERT, following the practice recommended in Hu et al. (2021). All remaining parameters in both encoders are frozen throughout fine-tuning, so only the LoRA adapter weights are updated during training. As a result, three designs were established for the experimental setup: image-encoder only, text-encoder only, and the multimodal encoder.

3.3 Modality Conditions

An ablation study was designed to evaluate the three modality conditions coherently. The image-only condition uses the ResNet-50 image tower alone, with the text encoder disabled. The classification head receives the 128-dimensional image vector as input. Beyond its role in the ablation study, this condition is also the most operationally realistic: radiology reports are generated after a radiologist has reviewed the image, so a multimodal model evaluated on paired image-report data benefits from a signal that would not be available at triage. The image-only model avoids this entirely.

The text-only condition runs Bio_ClinicalBERT on the radiology findings text, producing a 768-dimensional vector. The ResNet-50 image encoder is deactivated and the classification head receives only the text vector. To prevent label leakage, the findings text is masked to replace explicit pleural effusion terms with a generic placeholder, ensuring the model learns from descriptive visual language rather than the radiologist's diagnostic conclusion. Findings text coverage in the female subgroup is approximately 25%, meaning a substantial proportion of training examples have an empty text input. This makes the text-only condition particularly weak at low label budgets.

The multimodal condition has both encoders activated and their representations concatenated prior to the classification head, joining the 128-dimensional image vector with the 768-dimensional text vector to form an 896-dimensional joint vector. The head then predicts by drawing simultaneously on visual and textual evidence. This architectural design ensures that the three modality conditions differ only in which encoder pathways are active, while the head architecture and modality-specific LoRA configurations remain consistent across conditions. Table 1 in the next section demonstrates the different architectures and their respective hyperparameters.

4. Training Methods

4.1 Optimiser and Schedule

All models were trained using AdamW with decoupled weight decay of 1×10^{-4}. Separate learning rates were applied to the adaptation parameters and the classification head, as the head is randomly initialised which requires larger gradient updates early in training. Thus, LoRA matrices and prefix embeddings were trained at a learning rate of 2×10^{-4}, while the classification head was trained at a five-fold rate of 1×10^{-3}. Gradient norms were clipped to 1.0 to prevent instability at very low label budgets where individual batches can have high variance. Mixed precision training (AMP) was enabled on GPU to reduce memory usage and accelerate training. A fixed learning rate was used during training, with early stopping used to prevent overfitting instead of using a learning rate schedule with decay. A maximum of 8 training epochs was permitted per run.

4.2 Regularisation and Overfitting Mitigation

As LoRA is a PEFT technique, regularisation is automatically embedded in the text, image, and multimodal encoders. Rather than updating all model parameters, LoRA only updates a small subset of the parameters while freezing the rest at pretrained BioViL weights. For instance, the text encoder uses LoRA rank 8, meaning that it updates approximately 300k out of 110 million parameters from Bio_ClinicalBERT. This limits the model's ability to memorize training examples, even at high label scarcity.

Additional regularisation was implemented in all models through the classification head, which applies a 0.1 dropout before each linear layer. Early stopping was also used to end a seed run when three consecutive epochs produce worse validation AUROC than the current best.

Table 1: Hyperparameter settings used for training the text, image, and multimodal encoders. LR: Learning Rate

Hyperparameter	**Text Encoder**	**Image**	**Multimodal**
Optimiser	AdamW	AdamW	AdamW
Class. Head LR	1×10^{-3}	1×10^{-3}	1×10^{-3}
Encoder LR	2×10^{-4}	2×10^{-4}	2×10^{-4}
Batch size	16	16	16
Epochs	8	8	8
Dropout	0.05	0.1	0.05/0.1
Weight decay	1×10^{-4}	1×10^{-4}	1×10^{-4}

4.3 Text Input and Label Leakage

Training the text encoder requires appropriate text input, as it affects both performance and clinical usefulness of the model. Radiology reports in CheXpert Plus consist of distinct sections, namely the findings section, in which the radiologist describes objective observations from the image, and the impression section, where those observations are synthesised

into a clinical diagnosis. The binary pleural effusion labels in CheXpert Plus are extracted from the impression section of the reports using the CheXbert NLP labeller (Smit et al.). Using the impression section makes sense for labelling, as it requires a distinct diagnostic. However, training a model on impression text would simply teach it to replicate existing diagnostic judgement rather than classify new observations. This is not the intended use case, as an automated classifier should find clinical value by supporting decisions before or during radiological interpretation. Thus, we use the findings section as text input, which includes descriptive observations. Condition-specific terms that appear incidentally in the findings section (such as "small effusion noted") are additionally masked with the placeholder token to prevent label leakage during training.

4.4 Label Budget Protocol

To simulate clinically realistic label scarcity, all models were trained on fixed label budgets of N = {25, 50, 100, 200, 500, full}. Subsampling protocols differ across modalities. For the text encoder, subsampling was stratified to preserve the ∼67% positive rate of the full training set at every budget level. To account for the class imbalance in the female cohort, the text encoder applies a weighted binary cross-entropy loss that penalises misclassification of the minority class proportionally. For the image and multimodal encoders, subsampling was class-balanced (50% positive, 50% negative) at every budget to ensure both classes are sufficiently represented even at very small N. The image and multimodal encoders use unweighted binary cross-entropy, but subsampling at (50% positive, 50% negative) handles the imbalance here. Each modality has a separate development pipeline, but achieve imbalance-handling and the within-modality budget comparisons remain controlled. We expect the models to improve as the label budgets increase, however this was only observed marginally (Section 5).

To account for sampling noise, which is particularly severe at very low N, each budget was run across three fixed random seeds, giving 18 training runs in total. The validation and test sets were held fixed across all budgets and seeds to ensure comparability of metrics. The image-only and multimodal models additionally evaluated each run on three confound test sets: edema, consolidation, and pneumonia, to verify that performance gains at higher budgets reflect genuine pleural effusion sensitivity rather than visual correlations with similar conditions.

4.5 Implementation Notes and Computational Limitations

A single NVIDIA A100 GPU was used to run the experiments via Google Colab Pro. Before training could begin, we implemented a data pipeline to transfer the CheXpert Plus image dataset, totalling approximately 720GB, to a personal Azure Blob Storage and cached the images to the Colab runtime disk to save compute time and costs across runs. The cache had to be restored on subsequent runs because Colab enforces a strict 24-hour session limit, which proved to be a major constraint when running the full-budget training loop, especially for the multimodal condition.

Several optimisations were essential to fit within the available compute budget: pre-tokenisation of all text inputs, enabling mixed precision training (AMP), and clearing GPU memory between runs. These were applied to optimise GPU performance while staying within the compute budget. In addition, label budget granularity (six points from N=25 to full) was chosen to provide higher resolution in the low-data regime, where modality and budget effects are most pronounced, while balancing total compute cost across the three modality conditions and three random seeds. Total compute time for all three modality experiments and 18 runs spanned several Colab sessions, with full-budget runs requiring approximately 6 to 8 hours each, making the task computationally expensive.

5. Results

We evaluate the performance of text-only and image-only model configurations, each fine-tuned using parameter-efficient adaptation strategies across label budgets. We then compare baseline unimodal models to a multimodal fusion, evaluated primarily using AUROC values. AUROC measures how well the model separates positive from negative cases across all possible decision thresholds, rather than at a single fixed cutoff. This makes it robust to the class imbalance in our cohort (∼67% positive), as it evaluates the quality of predictions regardless of the relative frequency of each class (McDermott et al.).

Each of the three model configurations are trained and tested on the CheXpert Plus dataset (Chambon et al., 2024), restricted to female patients with binary pleural effusion labels. Preprocessing yields a cohort of 48,247 studies across 20,342 unique patients with a positive rate of approximately 67%, split into training, validation, and test sets as shown in Table 2. All results are reported as mean ± standard deviation over three random seeds $(7, 17, 27)$.

Table 2: Dataset statistics for the CheXpert Plus cohort after preprocessing. Uncertain (-1.0) and unmentioned (NaN) labels are excluded. All patients are female.

Split	N	PE+ (%)	Unique Patients
Train	38,712	26,235 (67.8%)	16,273
Validation	4,782	3,297 (68.9%)	2,034
Test	4,753	3,208 (67.5%)	2,035
Total	48,247	32,740 (67.9%)	20,342

To assess whether female-specific fine-tuning produces measurable improvements over general population training, a

text-only LoRA model was trained on the full mixed-sex CheXpert Plus cohort (108,386 studies) and evaluated on the same female-only test set used throughout this study. The baseline achieved AUROC 0.5290 ± 0.0012 and F1 0.8060 ± 0.0001 across three seeds (Seed 7: 0.5277, Seed 17: 0.5288, Seed 27: 0.5306).

5.1 Text Encoder

Table 4 reports test AUROC across all three text encoder adaptation strategies and six label budgets. All strategies perform at approximately random chance (AUROC 0.50) across the full label budget range, with marginal improvement observed only for LoRA at the full budget (0.5224 ± 0.0024).

The results indicate that the text encoder trained on the findings section with masking does not reliably learn to map radiological observations to pleural effusion labels under any of the tested adaptation strategies. Several factors may contribute to this, for instance the fact that CheXpert labels were derived from the impression section, while the model uses the findings section. As discussed in Section 4, this design was chosen to build a clinically meaningful pre-diagnosis classifier, but the consequence is a harder learning problem. Second, condition term masking removes the most discriminative features, requiring the model to infer the diagnosis from indirect clinical language. An effective model would need to learn the clinical relationship between observations and the diagnosis, which may require more labelled data than the budgets explored here.

5.2 Image Encoder

Test AUROC for the image-only LoRA condition remains near random chance across all six label budgets, with mean and standard deviation computed across three random seeds. Mean AUROC is 0.5000 at $N = 25$, 0.4947 at $N = 50$, 0.4927 at $N = 100$, 0.5024 at $N = 200$, 0.5109 at $N = 500$, and 0.5140 at full budget. No monotonic improvement with label budget is observed in the low-data regime; AUROC oscillates around 0.50 up to $N = 500$ before a marginal gain appears at full budget.

Seed-level variance is highest at $N = 200$, where test AUROC ranges from 0.483 to 0.521, suggesting sensitivity to the particular sampled examples rather than the emergence of a stable discriminative signal. At full budget, the model remains only weakly above chance, with AUROC 0.5140 ± 0.0089. Inspection of test-set positive rates shows that in many runs, especially at $N = 50$, $N = 100$, and $N = 500$, threshold selection leads the model to classify almost all test examples as positive. This threshold selection affects only threshold-dependent secondary metrics such as F1, precision, recall, and positive rate. It does not affect AUROC, which remains the primary metric used to compare image-only, text-only, and multimodal models. This produces recall near 1.0 and precision near the test-set base rate of approximately 0.675, so the observed F1 of approximately 0.806 mainly reflects majority-class prediction rather than meaningful discriminative performance.

At full budget, image-only LoRA achieves AUROC 0.5140, marginally below text-only LoRA at 0.5224 and the multimodal encoder at 0.523. This small gap suggests that the visual pathway acquires slightly less usable signal than the textual pathway, even when the full training set is available. This may reflect the greater representational shift required to adapt ResNet-50 image features to a demographically constrained female subgroup, despite using a higher LoRA rank of $r = 16$.

5.3 Multimodal Encoder

Table 4, in the appendix, displays the test AUROC recorded for the multimodal encoder across all six label budgets, with the mean and standard deviation reported across three seeds. Results show performance hovering near random chance across all budgets, with mean AUROC of 0.4963 at N=25, 0.4974 at N=50, 0.5002 at N=100, 0.4926 at N=200, 0.4992 at N=500, and 0.523 at full budget. AUROC oscillates around 0.49 until reaching 0.523 at full budget, which represents a meaningful improvement and tracks closely with the text-only encoder at full budget (0.5224).

Contrary to the central hypothesis, the multimodal fusion model does not produce disproportionate gains at low label budgets. It does not meaningfully surpass the other encoder configurations at any tested budget. The text input extracted from the findings section does not contribute an exploitable supervisory signal that complements the X-ray images. The masked findings text, which strips explicit pleural effusion terms and is present in only approximately 25% of the female training subset, is not a sufficiently strong supplementary data source for the multimodal encoder.

Table 3: Test AUROC at full label budget across all model configurations. Female cohort, CheXpert Plus, `section_findings` with condition masking unless stated otherwise. Results reported as mean ± std over 3 seeds.

Modality	**Strategy**	**AUROC (Full)**
Text-only (female)	LoRA	$0.5224_{\pm 0.0024}$
	Prompt Tuning	$0.4988_{\pm 0.0012}$
	LoRA + Prompt	$0.5034_{\pm 0.0048}$
Image-only (female)	LoRA	$0.5140_{\pm 0.0072}$
Multimodal (female)	LoRA	0.5230
Mixed-sex Baseline	LoRA (text-only)	$0.5290_{\pm 0.0012}$

Confound evaluation across the edema, consolidation, and pneumonia test sets shows similar near-chance performance. This suggests that the multimodal fusion model does not develop systematic visual associations with related thoracic conditions, and the marginal full-budget improvement does not

stem from spurious correlations with confounding pathologies.

6. Conclusion

Overall, we investigate parameter-efficient fine-tuning techniques as a method for improving the medical vision-language model BioViL on the female subpopulation in CheXpert Plus. The central hypothesis that multimodal fusion would provide disproportionately large gains at low label budgets was not supported by our results. All three model configurations (text-only, image-only, multimodal) produced AUROC scores near random chance across all label budgets. The marginal improvement observed at full budget (0.523) suggests that with sufficient labelled data, the model begins to extract weak signals from the inputs, but the gain remains too small to recover clinically useful classification performance within the demographic and methodological constraints of this study.

The combination of demographic restriction, label noise from the NLP tool CheXbert, and the masking of diagnostic terms appears to dominate over any potential multimodal benefit, even at full label budget. Removing the most discriminative textual features by condition term masking allows a clinically valid model, yet, also makes the task more difficult. Additionally, there are inherent complications that arise from biomedical data, as reports often refer to a patient's prior imaging studies, which the model cannot compare to. Since each image is processed in isolation, temporal language such as "effusion has decreased" is ambiguous. It describes a change relative to a prior examination that the model cannot compare against, rather than describing the state of the current image (Bannur et al., 2023). Beyond the temporal limitations of BioViL, subpopulation fine-tuning introduces a clear technical challenge. As (Yang et al., 2023) demonstrates, models minimising overall training loss tend to learn spurious correlations that can fail in subgroups. The limited female-only labels may have been insufficient to adapt the pretrained representations to physiological patterns specific to the subpopulation, and a larger dataset may avert this issue.

Despite having nearly three times more training data, the baseline achieved AUROC 0.5290 ± 0.0012 which is marginally above the female-only full budget result, but equally near random chance. This suggests the performance ceiling is not a function of training set size or demographic composition, but rather of the fundamental mismatch between the label source and the text input. The CheXpert Plus labels were generated by an automated NLP labeller which, even at its highest performing variant CheXbert, only achieves an F1 of 0.798 against radiologist annotations (Smit et al.). Models trained on such labels learn to predict what the NLP system would say rather than what the actual radiologist reports address.

An extension of this work would be to replace BioViL with its temporal successor, BioViL-T (Bannur et al., 2023). It explicitly incorporates prior images and reports during both training and fine-tuning using a CNN-Transformer hybrid multi-image encoder. As $\sim$40% of radiology reports in MIMIC-CXR, a dataset closely related to CheXpert Plus, refer to a prior examination (Bannur et al., 2023), this approach could address the temporal ambiguity in our current setup.

Overall, this study motivates the use of more targeted data curation, label quality improvements, and model architectures aware of temporal features to reliably classify medical data for population subgroups.

Statement of Individual Contributions

- **67674:** Configured the Azure Blob Storage pipeline, constructed the train/validation/test splits, and implemented and ran the baseline models including evaluation on mixed-sex cohorts.
- **65547:** Wrote the text-only encoder testing LoRA, prompt tuning, and combined fine-tuning. Ran the full budget sweep.
- **66190:** Wrote the image-only encoder model and fine-tuned using LoRA. Ran the full budget sweep.
- **73639:** Wrote the multimodal fusion model integrating both encoders and ran the full budget sweep.

Statement of AI Use

Claude (Anthropic, 2025) was used to check code, debug, and create LaTeX syntax for the final document.

Log Building LaTeX File

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Appendix

Table 4: Text Encoder test AUROC (mean ± std over 3 seeds) across label budgets and adaptation strategies. Female cohort, CheXpert Plus, `section_findings` with condition masking.

Strategy	**N=25**	**N=50**	**N=100**	**N=200**	**N=500**	**Full**
LoRA	$0.4995_{\pm 0.0012}$	$0.4977_{\pm 0.0028}$	$0.4998_{\pm 0.0051}$	$0.5053_{\pm 0.0108}$	$0.5055_{\pm 0.0007}$	$0.5224_{\pm 0.0024}$
Prompt Tuning	$0.5008_{\pm 0.0008}$	$0.5000_{\pm 0.0024}$	$0.5000_{\pm 0.0028}$	$0.5007_{\pm 0.0006}$	$0.4995_{\pm 0.0048}$	$0.4988_{\pm 0.0012}$
LoRA + Prompt	$0.5006_{\pm 0.0022}$	$0.4988_{\pm 0.0010}$	$0.4990_{\pm 0.0003}$	$0.4998_{\pm 0.0029}$	$0.5023_{\pm 0.0044}$	$0.5034_{\pm 0.0048}$

Table 5: Multimodal test AUROC (mean ± std over 3 seeds) across label budgets. Female cohort, CheXpert Plus, `section_findings` with condition masking.

Modality	**N=25**	**N=50**	**N=100**	**N=200**	**N=500**	**Full**
Multimodal	$0.4963_{\pm 0.0058}$	$0.4974_{\pm 0.0072}$	$0.5002_{\pm 0.0094}$	$0.4926_{\pm 0.0072}$	$0.4992_{\pm 0.0065}$	0.5230